Alex Hale's Board Map Legend

Oct 5, 2026 · @Alex

Quick start

Alex Hale's Board (called Joseph's Board until Oct 7, 2026; the folder and file names still say josephs) is a market dashboard plus two trading robots ("algos"), all running 24/7 on the spare Mac (Alexs-MBP) in the folder `~/josephs-board`. You control it from Terminal with the `./jb` command, and you look at it in a web browser.

The five things you'll use most:

1. `cd ~/josephs-board` first, every time you open Terminal.
2. `./jb status` to see if every part is running.
3. `./jb restart` to fix most problems. It is safe and loses nothing.
4. On your phone, open **http://100.93.57.24:8090/** (Tailscale must be on).
5. A green dot on a board tab means that part is running.

Where this legend came from: the code you uploaded (bridges, web, IBKR algo, lib, gateway), the two fixes installed on Oct 5 (delayed prices and the Webull fix), and your chat notes. Anything marked *(from notes)* came only from your notes, because that code wasn't in the upload. Every board tab also has a "? Legend" button that explains that tab on the spot, once the legend update is installed.

./jb commands

Type these in Terminal on the spare Mac after `cd ~/josephs-board`. The `jb` script itself wasn't in the upload, so the first six come from your notes and the rest from the algo's own instructions.

Command	What it does	When to use it
`./jb status`	Lists all 7 parts and whether each is running, plus the board addresses.	First thing whenever something looks off.
`./jb restart`	Stops and starts everything. Safe to run more than once; nothing is lost.	After installing new files, or when a part shows red.

Command	What it does	When to use it
`./jb restart <name>`	Restarts just one part, e.g. `./jb restart ibkr`.	When only one tab has a problem.
`./jb logs <name>`	Shows that part's live log. Names: ibkr, webull, scanner, gateway, webull-algo, ibkr-algo. Press **Control+C** to stop watching.	When asked for "the last few lines" of an error.
`./jb webull-login`	Checks Webull API access and, if needed, approves it.	If the Webull tab can't connect or Webull asks you to re-approve.
`./jb url`	Prints the board addresses (Mac, phone, work laptop).	When you forget a link.
`./jb backtest ibkr <strategy file>`	Tests a strategy on past prices. Never touches your account. Add `--duration "30 D"` for more history or `--sessions all` to include pre-market, after-hours and overnight.	Before paper or live trading a new idea.
`./jb backtest webull <strategy file>`	Same, for the Webull algo *(from notes)*.	Same.
`./jb live ibkr <strategy file>`	Starts **real-money** trading. Asks you to type `I_UNDERSTAND_THE_RISK`. Runs in that Terminal window; Control+C stops it and cancels any open order.	Only after backtests and paper trading look good.
`./jb live webull <strategy file>`	Same, for Webull *(from notes)*.	Same.
`./jb live-reset ibkr <strategy file>`	Clears the algo's own record of its position after you've fixed a mismatch with your real IBKR position.	Only when live trading refuses to start because the positions don't match.

The 7 services

Every part starts by itself when the Mac restarts and restarts itself if it crashes. Only the gateway (8090) can be reached from other devices; the rest listen on the Mac alone and the gateway passes their data through.

Name (for `./jb logs`)	Port	What it does	Board tab
gateway	8090	The front door: serves the board, asks for the password (once per device, remembered for a year), and passes live prices from the other parts to your browser.	All of them
ibkr	8765	Streams IBKR prices from IB Gateway: stocks, ETFs and futures. Falls back to free delayed prices for symbols you don't pay for. Records every price around the clock.	IBKR
webull	8766	Streams Webull prices for stocks and ETFs (no futures). Records every price around the clock.	Webull
scanner	8767	The mover scanner: watches *your ETF list or 30 big stocks for unusual moves*.	Scanner
webull-algo	none	The Webull trading robot (paper or live) *(from notes)*.	Webull Algo
ibkr-algo	none	The IBKR trading robot. Currently in **paper** mode.	IBKR Algo
keepawake	none	Stops the Mac from going to sleep so everything keeps running *(from notes)*.	none

After a Mac reboot you must log in to **IB Gateway** yourself; the IBKR parts wait for it. Tailscale starts on its own.

Addresses

All three open the same board and ask for the same password.

Device	Address	Needs
The spare Mac itself	http://localhost:8090/	Nothing
Your phone (and any device with Tailscale)	http://100.93.57.24:8090/	Tailscale app turned on
Work laptop (no Tailscale)	https://alexs-macbook-pro.tail60bf05.ts.net/	Tailscale Funnel on. Note the **s** in alexs. This link is on the public internet, so keep it private and the password strong.

Each board can also be opened alone by adding its page name, e.g. `.../ibkr.html` or `.../webull.html`. You can also "Add to Home Screen" on the phone for an app-like icon.

Funnel commands, typed on the Mac:

- Turn on: `/Applications/Tailscale.app/Contents/MacOS/Tailscale funnel --bg 8090` (stays on after reboots)
- Check: `/Applications/Tailscale.app/Contents/MacOS/Tailscale funnel status`
- Turn off: `/Applications/Tailscale.app/Contents/MacOS/Tailscale funnel --https=443 off`

Board tabs

The top bar has five tabs. The board remembers the last tab you opened, and a tab disappears if its part is switched off in settings.env.

Tab	What you see	What you can do
IBKR	Live prices from Interactive Brokers in two groups, US EQUITIES and FUTURES. Futures follow the front-month contract and roll over automatically.	Add a symbol (e.g. `AAPL` or `ES=F`), remove one with **×**, switch **List / Tiles**.
Webull	Live prices from Webull for stocks and ETFs. Futures tiles here say to use the IBKR board. Webull streams at most 100 symbols.	Same as IBKR.

Tab	What you see	What you can do
Scanner	Mover Scanner with ETFs | Stocks buttons: your ETF list, or 30 heavily traded stocks (list in mover-scanner/stocklist.yaml). ETFs flag at a 0.5% move in 5 minutes (1% in 15); stocks at 1% (2% in 15). Shows a sortable table (Symbol, Last, Window %) and a "Recent flags" list.	Switch ETFs / Stocks (back in about 10 seconds), click a column heading to sort.
Webull Algo	The Webull robot's status, last backtest and Trading options.	Change and save Trading options.
IBKR Algo	The IBKR robot: mode (PAPER or LIVE), strategy, last price, session now, position, unrealized P&L, return since start, last order, last backtest and Trading options.	Change and save Trading options.

On every price board:

- **Ticker tape** across the top scrolls every symbol with price and change.
- **Clock** shows the time and date; under it, **equities** and **futures** status lights say when the last update came in.
- **Change boxes**: 1M, 5M, 15M, 45M, 4H, 8H, 1D, 3D, 1W, 1MO, each the % change over that window. On a phone (List view), the headline number is the 1D change; tap a row to see all boxes.
- **Bottom line** of each tile: where the price came from (`ibkr · live`, `ibkr · delayed ~15m`, `webull · live`, `webull · overnight`) and how old it is.
- **Your symbol list is shared** across all your devices, and the Mac keeps recording prices around the clock, so overnight moves are there when you look.
- **Connection** button: leave it blank. It's only for unusual direct connections.

Colour, tag and box legend

You see	Where	It means
Green dot	Tab name	That part is running.

You see	Where	It means
Red dot	Tab name	That part is not running on the Mac. Run `./jb status`.
Grey dot	Algo tab name	The robot isn't running, but old backtest results are still shown.
Green / red price number	Tile	Price is up / down over the last **1 minute**.
Green / red tint and left edge	Tile background	Price is up / down over the last **2 minutes**.
Quick green or red flash	Tile	The price just changed.
Green / red change box	1M to 1MO boxes	That window's change is up / down.
"…" in a change box	Change boxes	Not enough history yet to cover that window. It fills in by itself.
Green status light, "last update Xs ago"	Under the clock	Prices are flowing.
Red status light	Under the clock	No update for 45+ seconds, or the connection dropped.
Yellow **DELAYED** tag, price slightly faded	Next to a symbol (IBKR)	You don't pay for live data on this symbol (e.g. CL=F, GC=F), so it shows IBKR's free prices, about 10 to 20 minutes old. The algo never trades on these.
DLY	Ticker tape	Same as DELAYED.
Yellow box: "No overnight prices here…"	Top of Webull board	Your Webull plan doesn't include the overnight session (8 pm to 4 am). Regular and extended hours still work.
"no Webull futures data subscription -- see this symbol on the IBKR board"	Futures tile on Webull board	Normal. Futures come from IBKR.

You see	Where	It means
Red box at the top of a board	Price board	A real problem, e.g. "The IBKR bridge is not running on the spare Mac". It tells you which command to run.
Red text on a tile's bottom line	Tile	That one symbol has a problem (e.g. unknown symbol).
Yellow note about 100 symbols	Webull board	You've added more than Webull will stream on one App Key.
Amber row with **FLAG**	Scanner table	That ETF made an unusual move in the last 5 minutes.
Coloured lines in "Recent flags"	Scanner	Which detector fired: blue = price jump, amber = volume surge, purple = range expansion, teal = buying/selling pressure.
PAPER / **LIVE** tag	Algo tab	Whether the robot is simulating or trading real money.
Red **LIVE TRADING PAUSED** banner	Algo tab	The robot stopped itself because an order's outcome couldn't be confirmed. Check the broker app before restarting it.

Trading algos

Each robot follows a **strategy file** (a short text file of rules, e.g. "buy when the 20-minute average crosses above the 50-minute average"). The IBKR and Webull robots use the same file format, so a strategy works with either. The IBKR robot is in **paper** mode today.

The three modes, safest first

1. **Backtest**: replays past prices through the rules. Pure arithmetic; your account is never touched. Results appear on the algo tab.
2. **Paper**: runs on live prices but every order is simulated on the Mac. Nothing is sent to the broker.
3. **Live**: real orders, real money. Needs two separate switches (below).

Example strategies included (IBKR, in `ibkr-algo/strategies/`): `sma_crossover`, `rsi_reversal`, `bollinger_breakout`. Strategies trade plain US stocks and ETFs only, never futures or options.

Trading options panel (on each algo tab; changes take effect within about 5 seconds)

Setting	Default	What it does
Sessions	Regular hours only	Regular hours (9:30 am to 4:00 pm ET) are always on. Tick pre-market (4:00 to 9:30 am), after-hours (4:00 to 8:00 pm) and overnight (8:00 pm to 3:50 am, Sun to Thu nights) separately for **Paper & backtest** and for **Live**.
Order type	Limit	Limit or Market. Market orders go out only 9:30 am to 4:00 pm; outside those hours a limit order is sent instead.
Limit price	0.05%	How far from the last price to set the limit so it fills quickly. A negative number waits for a better price (entries only).
Cancel if not filled after	60 seconds	An unfilled order is cancelled and shown as "unfilled", not an error. The robot tries again if the rule still holds.
Trailing stop	Off	Exits when the price falls this % from its best since entry. Watched by the robot, not held at the broker, so it only works while the robot runs.
Extra cost outside regular hours	0.1% per fill	Paper and backtests only, to account for wider spreads.
Use the strategy file's settings		Clears the panel so the strategy file (or built-in defaults) applies again.

Which setting wins: the panel first, then the strategy file's `trading:` section, then built-in defaults.

Safety rules (live trading)

- Two switches must both be on: `confirm_live: true` plus a maximum order value in the strategy file, **and** typing `I_UNDERSTAND_THE_RISK` when you run `./jb live`.
- Every order is checked against a share limit and the maximum order value.
- It refuses to start if your real position differs from its own record, or an order is still open.
- One order at a time; only what the broker reports as filled is counted.
- A rejected order means no new order for 15 minutes; 3 rejections in a row stop it.
- If it can't confirm what happened to an order, it pauses and shows a red banner.
- It never trades on delayed (yellow DELAYED) prices, in paper or live mode, and says so on the IBKR Algo tab.
- Control+C in the live window stops it and cancels any open order.
- Start small, watch the first trades, and always check fills in the broker's own app.

settings.env switches

`~/josephs-board/settings.env` holds your private keys and on/off switches. Only the names are listed here. Never paste this file into a chat, and run `./jb restart` after changing it.

Name	What it controls
`JB_PASSWORD`	The board password (secret).
`WEBULL_APP_KEY`, `WEBULL_APP_SECRET`	Your Webull API keys for the real account (secret).
`RUN_WEBULL_BRIDGE`	Turns the Webull price feed on or off *(from notes)*.
`RUN_WEBULL_ALGO`	Turns the Webull robot on or off *(from notes)*.
`WEBULL_ALGO_STRATEGY`	Which strategy file the Webull robot runs *(from notes)*.
`RUN_IBKR_ALGO`	Turns the IBKR robot (paper) on or off.
`IBKR_ALGO_STRATEGY`	Which strategy file the IBKR robot runs, e.g. `strategies/sma_crossover.yaml`.

Name	What it controls
`IBKR_ACCOUNT_ID`	Only needed if one IBKR login has several accounts.
`IBKR_GATEWAY_PORT`	The port IB Gateway listens on. Leave as set up.
`WEBULL_ENVIRONMENT`, `WEBULL_REGION_ID`	Which Webull system and region to use. Leave as set up.

Not a setting to put in the file: `IBKR_LIVE_TRADING_CONFIRM` is set only for one Terminal window when you type the confirmation in `./jb live`. Other names you may see (bridge tokens, data folder, bind addresses) are internal plumbing; leave them alone.

Folders, logs and backups

Everything lives in `~/josephs-board` on the spare Mac.

Folder or file	What's in it
`jb`	The control command.
`settings.env`	Keys, password and switches (private).
`install.sh`, `requirements.txt`	The installer and its list of parts. Already run.
`gateway/`	The front door (port 8090).
`bridges/`	The IBKR and Webull price feeds.
`mover-scanner/`	The Scanner.
`webull-algo/`, `ibkr-algo/`	The two robots, their `strategies/` folders, `logs/`, and `trading_options.json` once you save the panel. Backtest trade logs and charts land next to the strategy file.
`web/`	The board pages your browser shows.
`lib/`, `bin/`	Shared helpers and service starters.
`data/`	Your shared symbol list and the price history the Mac records (about 35 days).
`logs/`	Each service's log, read with `./jb logs <name>`.

Folder or file	What's in it
`backup-delayed/`	Your files from before the delayed-prices fix (Oct 5).
`backup-webull/`	Your files from before the Webull overnight/futures fix (Oct 5).

To undo the delayed-prices fix: `cd ~/josephs-board && cp backup-delayed/ibkr_bridge.py bridges/ && cp backup-delayed/board.js backup-delayed/board.css web/ && cp backup-delayed/paper_trader.py ibkr-algo/algo/ && ./jb restart`

In this project's files: `JOSEPHS_BOARD_v3.zip` (the original full code), `Josephs_Board_Setup_Guide.pdf` (setup Steps 1 to 11), and `josephs-board/` (the Oct 5 fixes and their steps).

Overnight features (October 2026 updates)

Everything below was added in five updates. Each page also explains it under **? Legend**.

Feature	Where	What it does
VS 4PM CLOSE · OVERNIGHT · 9:30 GAP boxes	IBKR and Webull boards	Change since the 4 pm close, the 8 pm – 4 am overnight move, and today's opening gap. The 1D box is now called **24H** (the last 24 hours).
O/N range bar, 1W and 52W bars	Boards	Overnight low to high with a dot at the price now. Green dot = above the overnight high, red = below the low. Under it, two smaller bars: 1W (last 7 days) and 52W (last year), low to high with a dot at the price now, counting after-hours and overnight too. Green dot = at a new high for that range, red = at a new low. Hover or long-press for the dates.

Feature	Where	What it does
Session badge	Under the clock	Which session it is in New York and how long until the next one (holidays aren't known).
Ticker tape futures	Top of every board	ES, NQ, YM grouped after a yellow FUTURES label with their change vs. close (Webull gets them from IBKR).
A–Z order, faded Webull tiles	Boards	Symbols listed alphabetically. Webull stock tiles fade 8 pm – 4 am because the Webull plan has no overnight prices.
US EQUITIES ▾ menu	Boards	Show only holding, with alerts, or moving overnight ±1%. Nothing is removed.
Alerts	Boards, Alerts button	Price above/below or % move vs. close, checked by the Mac day and night, pushed to the phone through the free ntfy app.
Morning cards	Top of the boards, phone	7:00 and 9:00 am ET weekdays: futures, biggest moves, overnight high/low breaks. Copy for Claude button.
Recap video	Recap tab; sent to phone (ntfy) at 8:05 am ET on trading days	2-3 minute narrated video since the 4 pm close: after-hours (SMART), overnight (Blue Ocean), pre-market (SMART), then where things stand with your holdings. Make one now button; time set by JB_RECAP_TIME in settings.env (or off). Last 14 kept in data/recaps/.
Chart lines and markers	Price chart pop-out	4 pm close (yellow), overnight high/low (purple), news ▼, Compare line, volume bars, your cost (pink), your buys and sells.
Since 4 pm close, bold keywords	News tab	Only news after the last close; words like earnings or FDA in bold yellow.

Feature	Where	What it does
Vs close column, session chips, volume ratios	Scanner, Movers	Change vs. close, flags filtered by session, and volume vs. this session and vs. the last hour.
Overnight picks, earnings, track record	Scanner, Picks	Overnight setups, next earnings dates (Nasdaq), news in the last hour, and a daily scored pick history.
Portfolio tab	Portfolio	Type in each buy and sell; positions with average cost and gain/loss; HELD line on the boards; basket backtest vs. SPY (Yahoo daily prices).
Overnight gap strategy	IBKR Algo (paper)	Buys 7:30 – 8 pm ET Mon–Thu if up 0.5%+ vs. close, sells at the 9:30 open; 2% stop, 4% take-profit. File: ibkr-algo/strategies/overnight_gap.yaml.
Run backtest button	Both algo tabs	Pick a strategy and run a backtest on the last 10 days of 1-minute prices, no Terminal.
Paper trading report	Both algo tabs	One line per finished trade with sessions, entry, exit, P&L and running total, grouped by day, with a line chart.
Stop & close position	Both algo tabs	Kill switch: closes the position and stops new trades until Resume trading. Holds through restarts.
Webull auto-reconnect	Webull bridge	Reconnects by itself after the 417 INVALID_SESSION error.

New files and data: gateway/watch.py, portfolio.py, algoctl.py; data/alerts.json, morning.json, holdings.json, ibkr_volume.json, webull_volume.json, picks_history.json, earnings.json; JB_NTFY_TOPIC in settings.env. Backups of each update are in folders named backup-overnight, backup-watch, backup-scanner2, backup-portfolio and backup-algo.

World, Signals, Analysis, My trading and Extras

Five tabs and extras added on Oct 6, all free sources. Each tab's **? Legend** explains it on the page.

Feature	Where	What it does
World tab	After Webull	Overnight mood meter (-100 to +100), market clock, futures, VIX, rates, dollar, Asia and Europe, crypto, economic calendar (Fed, data, Treasury auctions). Sunday evening push when futures open. Prices from Yahoo.
Signals tab	After News	Phone-alert switches: big or fast move, heavy volume, yesterday high/low, 4 am gap list, StockTwits buzz, SEC 8-K / insider buy / 13D, earnings move, 7:30 am overnight summary.
WHY? button, EARNINGS / BUZZ / 8-K tags	Board tiles	WHY? lists the likely reasons a stock is moving (news, filings, earnings).
Analysis tab	After Signals	Gap scorecard (how often gaps fill), session split (after-hours / overnight / pre-market / regular), sector heatmap, relative strength vs SPY and QQQ, stocks that move together, options max pain and put/call ratio.
Chart indicators	Price chart pop-out	VWAP, MA 20, MA 50, RSI, previous day high/low buttons.
IBKR 366 note	Under board tiles	The "No historical data query found (366)" message now says why it is harmless.

Feature	Where	What it does
My trading tab	After Analysis	Journal (IBKR fills every 5 min, Webull every 10, or by hand) with tags and notes; win/loss by session, tag and stock; daily totals; size calculator; daily loss limit push with optional algo pause; paper algo leaderboard (8:30 pm ET); replay a recorded night.
Weekly recap video	Recap tab, phone	Fridays 8:15 pm ET (JB_WEEKLY_TIME), or Make weekly one now: week's best and worst, market weather, Picks and your trades, best paper strategy, next week's earnings and big events. Last 8 kept.
Market weather	Recaps, phone page	Each sector fund gets sunny / partly sunny / cloudy / rainy / stormy from its move since the close (2.5x wider steps for the week).
Health tab	Last tab	What is running, when each broker last sent a price, disk space, last log lines, and a Restart button per part (same as ./jb restart).
Phone page	Phone link, top bar	One screen: mood, futures, weather, biggest moves, holdings, today's trading, alerts, coming up. Safari: Share, Add to Home Screen.

Feature	Where	What it does
Look button: Polished / Pure black / Clean light / Night / Auto	Button, top right	Tap it (top right of the board and the phone page) for a list. Polished is the normal look: calmer green and red, rounded tiles with a little space between them on a coloured area, AH badge, clean font for labels and buttons. Pure black and Clean light (grey and white) are polished too. Night = dim red, on every page, and the phone's bottom bar stays. (Classic and Old light were taken out on Oct 9: they copied Polished and Clean light.) Auto = Night 9 pm to 6 am, Polished otherwise. Only colours change, never what the board does. Remembered per phone or computer. If tiles look cramped on the Mac, press Command+0 (Safari zoomed out).

New files and data: gateway/world.py, signals.py, analysis.py, trading.py, extras.py; web/world.html, signals.html, analysis.html, trading.html, health.html, phone.html, theme.js; data/world.json, signals_log.json, analysis_cache.json, gap_log.json, journal.json, leaderboard.json, weather.json. Backups: backup-world, backup-signals, backup-analysis, backup-trading, backup-trading-fix, backup-extras.

If a part shows red, open the **Health** tab and press Restart before using Terminal.

SELL ALL, phone fixes and futures pop-up

Added on Oct 6. SELL ALL is an emergency button that sells everything on one broker; it passed a live test on both IBKR and Webull (1 AGNC share each). Nothing else on the board was removed.

Feature	Where	What it does
SELL ALL button	IBKR and Webull boards, next to Connection; Emergency card on the Phone page	Opens a window showing the account (paper or LIVE), every position and open order, and how it will sell now. Nothing is sent until you type SELL ALL and press the red button.
What SELL ALL does	That broker only	Pauses that broker's algo, cancels every open order, then closes every stock (and IBKR futures). Options are left alone. You get a phone push when it finishes.
How it sells	By session	Regular hours: market orders. Pre-market, after-hours, overnight: limit orders 1% through the last price, then 2%, then 3%, 20 seconds each. Markets closed: cancels and pauses only.
After SELL ALL	Algo tabs	The algo stays paused until you press Resume on its tab.
Webull account	settings.env	WEBULL_ACCOUNT_ID picks the Individual Margin account (your keys also see an Events Cash account).
Webull speed limit	Automatic	If Webull says "too many requests", SELL ALL waits a few seconds and asks again (jb-killfix).
Tabs list	Phone, top-left of the tab bar	The ☰ Tabs button lists every tab, so you don't have to scroll sideways.
Phone fixes	Every page on a phone	Typing no longer zooms the iPhone in; buttons and checkboxes are bigger; board header text is larger; wide tables keep the symbol column in place, with a "swipe sideways for more" hint. Computer screens are unchanged.

Feature	Where	What it does
Futures pop-up	Phone page, Futures and fear tiles	Tap a tile to see the price, change vs the 4 pm close, since yesterday, the 24-hour high and low, and a 24-hour chart.

New files on the Mac: gateway/killswitch.py, web/kill.js, web/mobile.js, web/phonefut.js. Backups are in backup-kill/, backup-phonefut/ and backup-killfix/.

Final ideas (Oct 7): orders, ETF edge, feeds, guardian, tools

Ideas 1-72 from the final-ideas list (all except 21), installed with jb-ideas. Most parts are new sections at the bottom of the **Night desk**; nothing else on the page moved. Night desk guide part 2 (pages 9 onward of the guide PDF) has the details.

Part	Where	What it does
Safety stops, targets, trailing, break-even	Night desk; ticket **More**	The board watches the price in every session and sells (stepping 0.5% to 3% under the price) when hit. Watched by the board, not placed at IBKR
Ticket extras	Order ticket	Good for: session / overnight + tomorrow / until cancelled; ladder; dollar size buttons; IBKR preview; duplicate warning; Flip; Chase
Scheduled orders	Night desk	At the open, 4 am, 4 pm, 8 pm or a set time, with optional conditions; paper by default; sell all at the open
Fill pushes and notes	Phone; My trading	A push per fill with the distance from your limit or last price; an automatic note per fill; scorecard; weekly PDF report Friday 8:30 pm

Part	Where	What it does
Second chip row	IBKR and Webull boards	Trend light, FAIR, BUYERS / SELLERS, PRICES DISAGREE, REDDIT #, NOTE, the fund card
Options lines	QQQ / SPY charts	Options magnet, fast and gamma flip levels as dotted lines
World additions	World tab	VIX curve warning, Fed meeting odds, Nasdaq-100 breadth, Bitcoin order book (Gemini)
3x ETF edge	Night desk	Fair value, rebalance, decay, drift, opex calendar, net liquidity, Treasury auctions, CFTC, yen and VIX pushes, price history
Feeds	Night desk	Top news ranked, IBKR newswire, Reddit, Bluesky, Polymarket, policy posts, buy / sell pressure, IBKR vs Webull cross-check
Mac guardian	Night desk; System check; Health	Power-cut restart, memory, clock, macOS updates, crash loops, backups 2:30 am, internet drops, SELL ALL re-check, Sunday self-test, Saturday refresh
Algo safety	Both algos	Heartbeat, skip entries when prices disagree; live only: position check, safe mode, order speed limit
Command bar and keys	Top bar ⌕ or /; Phone search box	TQQQ, TQQQ buy, TQQQ news, flip SOXL, tab names; keys 1-9, 0, B, S, Esc, ?
Any-symbol chart	Command bar	Any symbol, future or index in the board's own chart (Yahoo prices)

Part	Where	What it does
Filled-in chart	Charts	Stretches filled from IBKR after downtime, drawn in a dimmer blue-grey
Notes, what changed, widget, Siri, alert manager, go-live	Night desk; board and Phone	Symbol notes, a since-you-last-looked box, iPhone widget and Siri shortcut, every alert on one page, go-live checklist (./jb live runs it)

New files in ~/josephs-board: gateway/orders.py, edge.py, feeds.py, guardian.py, tools.py, iblink.py; bin/golive.py; web/nd-orders.js, nd-edge.js, nd-feeds.js, nd-guardian.js, nd-tools.js, plus2.js, world-plus.js, tools-ui.js, trading-plus.js, health-plus.js. New folders: backups/ and web/reports/. Backup of the old files: backup-jb-ideas/.

Sentiment report and phone link boxes

Added on Oct 7 (jb-sentiment). Both are explained in the board's **? Legend**.

Where	What it does
MACRO tab: Market sentiment report	**Generate report** makes one report on the market's mood by fixed rules from free data the board already has. Short term (next 1-3 days): overnight mood meter, VIX level and move, VIX curve, last 5 days, today's breadth, 2 days of macro headlines. Mid term (next 2-8 weeks): S&P and Nasdaq vs their 50-day average, last 20 days, Nasdaq-100 stocks above their 50-day, Fed odds, hedge fund positioning, a week of headlines, net liquidity. Long term (next 3-12 months): the MACRO score, S&P vs its 200-day average, 50-day vs 200-day, 52-week range, last 12 months, 6 months of net liquidity.

Where	What it does
Scores	Each line scores -1 (bearish) to +1 (bullish); a time frame is their average x 100; the combined score is the three averaged equally. Bullish +40 or more, Leaning bullish +15, Neutral, Leaning bearish -15, Bearish -40 or less. Also: one sentence on how the time frames fit together, a line for 3x bull vs 3x inverse funds, and big events in the next 3 days. Rules, not advice.
Report buttons	Copy text; Print / Save as PDF (prints just the report); Ask Claude about this report; Earlier reports lists the last 10 (60 kept on the Mac).
Phone page: link boxes	All ten bottom links as equal cyan boxes, two per row: Night Desk, Order Ticket, IBKR Algo, Webull Algo, Full Board, Macro, Ask Claude, Recap Video, System Check, Health.

New files: gateway/sentiment.py, web/sentiment.js; web/phone-links.js replaced. Backup in backup-jb-sentiment/.

If something looks wrong

Try these in order. If one shows an error, send only its last few lines.

What you see	Try this
A red dot or red box on a board	`./jb status`, then `./jb restart`.
IBKR tab red after the Mac restarted	Log in to IB Gateway on the Mac, then wait a minute.
Webull tab can't connect	`./jb webull-login`.
Board asks for a password again	Normal on a new device or after clearing the browser. Enter the board password.
Phone can't open the board	Check the Tailscale app is on, on both the phone and the Mac.

What you see	Try this
Work laptop can't open the link	Check the **s** in alexs, then run the Funnel status command (Addresses section). If it's off, run the turn-on command.
A tile shows DELAYED	Normal for symbols you don't pay for. Buy the IBKR data and run `./jb restart` to make it live.
Change boxes show "…"	Wait; they fill in as history builds.
Algo tab shows LIVE TRADING PAUSED	Check your position and orders in the broker's app before doing anything else.
Anything else	`./jb logs <name>` for the part with the problem, and send the last few lines.

Phone bar, Chart and Report Card (jb-reportcard)

Three additions from October 9, 2026: easier page moves on the phone, a chart for any ticker, and a Report Card where the board grades its own calls.

Feature	Where	What it does
Phone bar	Bottom of every page on a phone	**Back** (the page or tab you came from; on the full board it steps back through the tabs you opened), **Home** (the Phone page), **Chart**, **Menu** (every page and tab as big boxes). Computers are unchanged.

Feature	Where	What it does
Chart tab	Full board tab, and Chart in the phone bar	Type any ticker (TQQQ, AAPL, NQ=F, ^N225) for its line chart over 1D, 5D, 1M, 6M, 1Y, 5Y or All (its full history, one point a month). Prices from IBKR first, including the overnight venue; Yahoo when IBKR can't (indexes, crypto, stocks your IBKR plan doesn't cover). Shaded with a paler line = outside regular hours; cyan dashed = 9:30 am open; grey dashed = price at the start of the range. Your last 8 symbols are buttons.
Chart order flow	Chart tab at 1D, the Order flow buttons	Four extras, each with its own on/off button (turn all off for a clean chart). Buy/Sell line: shares bought at the ask minus shares sold at the bid, added up since 8 pm ET (dashed part = estimated from 1-minute bars before the page started watching). Big trades: 10x the day's average size and worth $100k+, circled on the line and listed. Volume at price: side bars of shares traded at each price today, busiest in amber. Bid/Ask: the best bid and ask now with shares on each side. Free IBKR data, so only symbols with live IBKR prices, up to 3 at a time; works pre-market, regular, after-hours and overnight. Not the full order book (that needs paid Level 2).

Feature	Where	What it does
Report Card tab	Full board tab, and Report Card in the phone menu	How often each feature was right: mood meters, Picks and each of their signals, ETF pick, gap scorecard, 4 am gaps, alerts, sentiment report, MACRO score, earnings expected moves, max pain, put/call, politicians' trades, relative strength, market weather and event days. Green 55%+ right, amber 45-55%, red under 45%, grey = fewer than 20 graded. 50% = a coin flip.
Live vs Replay	Each Report Card card	Live = calls saved since install, graded with free Yahoo prices 30 and 60 minutes later, at the next open and close, and 1, 5, 20 and 60 days later. Replay = the same rules re-run on up to 2 years of past prices, for price-only features; it runs each evening and takes a few minutes.
Card details	Tap a card	Each part (meter window or signal), each session the call was made in, a line of accuracy over time, and the latest calls with a check or cross.
Your trades vs the board	Bottom of the Report Card	Win rate and average result of your journal trades when the Futures mood agreed with you, was against you, or had no clear call. Inverse ETFs count the other way round.
right 61% tags	Next to the Futures mood, Picks mood and Mood since the 4 pm close meters	How often that meter, over the window you picked, pointed the right way by the next close (the open for the 4 pm mood), with how many readings. Tap to open the Report Card.

Feature	Where	What it does
Weekly push	Phone, Fridays 8:30 pm New York	The week's best and worst feature. Switch it off at the bottom of the Report Card.

Roku TV view

Added on Oct 9: a read-only TV app called **Alex Hale's Board** for the Roku, sideloaded with Roku's free developer mode. The board's **? Legend** explains it too.

Where	What it does
Top row	Market mood, Futures mood and Picks mood meters (since the 9:30 open during regular hours, since the 4 pm close otherwise).
Middle: IBKR tiles	The IBKR board's stocks, then its futures, in two labelled groups, as tiles like the IBKR board: they flash green or red as the price moves, the tint and side bar follow the last 2 minutes, boxes show the same moves as the IBKR board (1M, 5M, 15M, 45M, 4H, 8H on one row; 1D, 3D, 1W, 1MO, O/N on the next), and the O/N, 1W and 52W range sliders sit under each. Prices every 2 seconds, up to 24 tiles, in the IBKR board's own order.
Bottom	Top movers, a Futures 24 h chart (one coloured line per IBKR future showing its % change over the last 24 hours, with a key listing them best first), Picks and Coming up (big reports plus the latest alerts) as lines of text, then the market weather across the full width. The rest refreshes every 15 seconds.
Read only	The TV can only look: no trading, SELL ALL, algo pause or settings, and no account money or positions on screen. It uses its own TV key (data/tv_key.txt), not the board password, and that key opens only the TV summary.

Where	What it does
Where it works	Home Wi-Fi first, then the ts.net link (any Wi-Fi). The top line says which one it is using.
Open it	Press Home on the Roku remote and pick Alex Hale's Board (at the end of the channels; move it with the star button, Move channel).
OK button	Reconnect now. It only asks the board again and changes nothing. If the Mac's Wi-Fi address changes, the TV learns the new one by itself through ts.net.
Last resort	If it still can't reach the Mac: check the Mac is on and awake, then run `bash ~/josephs-board/roku/make-tv-app.sh` and send the new Board-TV.zip from Downloads to the Roku (Roku at 192.168.1.145, user rokudev).
Lock out the TV	Delete data/tv_key.txt, restart the gateway, and make the TV app again.

New files: gateway/tvview.py (the TV feed, /api/tv), roku/app (the Roku app), roku/make-tv-app.sh (makes Board-TV.zip). Backup: backup-jb-roku/.

Alex's channel

Added on Oct 9: your board as a TV news channel, with an anchor's voice reading what is on screen. It plays on the Roku, the phone and the Mac. The board's **? Legend** explains it too.

Where	What it does
The show	Segments in turn: BREAKING (when a board alert fires), the opening with your three mood meters, futures with a chart that draws itself, Top movers, Picks with their reasons, news for your symbols, sectors, Coming up and the market weather. A ticker crawls along the bottom, and the corner shows ES, NQ and the clock.

Where	What it does
The anchor	The Mac's own free voice (the Recap voice, Samantha unless changed) reads each segment while it is on screen, so the voice always matches the picture. When the sound is off, the same words show as closed captions near the bottom of the screen, two lines at a time. A fresh show is made about every 2 1/2 minutes while someone is watching, and none at all when nobody is.
On the TV	In the Roku app, the right arrow opens the channel and the left arrow (or Back) returns to the board. In the channel, OK or Play turns the sound off and on, and the right arrow skips to the next segment.
On the phone and Mac	Click the small ▶ Channel button next to the look button (top right of the full board and the Phone page), open channel.html on the board, tap Watch Alex's Channel on the Phone page, or pick Alex's Channel in the phone Menu. Tap Tune in with sound, since phones only play sound after a tap. Turn the phone sideways for a bigger picture.
Read only	It only reads what the board already shows. No account money or positions are shown or spoken. The TV uses its TV key and the phone and Mac use the board login.
Voice settings	JB_CHANNEL_VOICE and JB_CHANNEL_RATE in settings.env give the channel its own voice. Otherwise it uses JB_RECAP_VOICE and JB_RECAP_RATE. Sound files are kept in data/channel/ and old ones are deleted by themselves.

New files: gateway/channel.py, web/channel.html, web/channel-link.js, and the Roku app's ChannelView.brs and ChannelTask files.